

Wearable device requirements

Project: ****

2025-07-24

Preface

This document was generated with the wearable device specification tool, that enables a researcher to gather important requirements for a wearable device in a study, so that the device can be selected based on the requirements. In some situations, a device might not be able to meet all the requirements, and the researcher might need to make a decision on which requirements are most important. This document is a resource to guide the decision process between researchers and manufacturers.

Restore this configuration in the web-tool¹

General information

Type(s) of study: unspecified

Required devices: unspecified

Description of the intended main use case(s):

Hardware requirements

Housing & design

Aspect	Requirement	Device specs
Mounting	unspecified	
Housing material	unspecified	
Max dimensions (WxBxH)	5 x 5 x 2.5 cm	
Maximum weight (grams)	unspecified g	
Special design considerations		

¹Note that depending on the size of text inputs, restoring the bookmark may not work properly. The link will always contain all entered data, but url input length can be limited by the browser. To avoid silently dropped form-entries upon restoring the link, use a browser that supports long url inputs (e.g., Chrome) and be succinct with free-form text. Further, if the automatic redirect to the browser does not work when left-clicking on the link, copy the link via the right-click context menu and paste it manually into your browser of choice.

Measurement & sensors

Aspect	Requirement	Device specs
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Battery & storage

Aspect	Requirement	Device specs
Battery: continuous recording runtime without recharge	≥ 2 weeks (with 10s recording interval & unspecified Hz sampling)	
Storage: minimum on-device capacity	unspecified observations ($\geq$ unspecified days with 10s recording interval)	

Operating environment

Aspect	Requirement	Device specs
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Controls & indicators

Aspect	Requirement	Device specs
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Connectivity & I/O

Aspect	Requirement	Device specs
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Other hardware considerations

Data requirements

Aspect	Requirement	Device specs
Minimum recording interval	≤ 30 seconds	
Flexibility of recording interval	unspecified	

Other requirements

Aspect	Requirement	Device specs
Available budget per device, including shipping	unspecified	